

Delhi Public School

Subject: Biology | Grade: 12th | Class: 12th

Time: 2 Hours

Maximum Marks: 25

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Answer all multiple-choice and true/false questions. Each question carries 1 mark.

Q1. What is the expected phenotypic ratio of the F₂ generation in a standard Mendelian monohybrid cross? [1]

[Easy]

- a. 3:1
- b. 1:2:1
- c. 9:3:3:1
- d. 1:1

Q2. When a red-flowered Snapdragon plant (RR) is crossed with a white-flowered Snapdragon (rr), the F₁ generation exhibits pink flowers (Rr). This biological phenomenon is known as: [1]

[Moderate]

- a. Codominance
- b. Incomplete Dominance
- c. Complete Dominance
- d. Epistasis

Q3. An individual with the AB blood group exhibits codominance because: [1]

[Moderate]

- a. Allele I^A is fully dominant over allele I^B
- b. Allele I^B is fully dominant over allele I^A
- c. Both alleles I^A and I^B express themselves equally and simultaneously
- d. Both alleles are recessive to the i allele

Q4. In a stable population following Hardy-Weinberg equilibrium, the frequency of a recessive allele (q) is 0.4. What is the frequency of the heterozygous genotype (2pq) in this population? [1]

[Hard]

- a. 0.16
- b. 0.48
- c. 0.36
- d. 0.60

Q5. According to Charles Darwin, the primary driving force behind organic evolution is: [1]

[Easy]

- a. Interspecific hybridization
- b. Survival of the fittest through natural selection
- c. Inheritance of acquired characters
- d. Sudden, single-step large mutations

Q6. Which Mendelian law cannot be formulated or demonstrated using a monohybrid cross alone? [1]

[Moderate]

- a. Law of Dominance
- b. Law of Segregation
- c. Law of Independent Assortment
- d. Law of Purity of Gametes

Q7. Gregor Mendel conducted his historical hybridization experiments on the garden pea (*Pisum sativum*) for a continuous duration of seven years (1856-1863). [1]

[Easy]

Q8. The inheritance pattern of human ABO blood grouping is a classic example of both multiple allelism and codominance. [1]

[Easy]

Q9. Analogous organs are anatomical structures that perform similar functions and indicate a shared common ancestry via divergent evolution. [1]

[Moderate]

Q10. The evolutionary mechanism of genetic drift operates with high intensity and produces significant evolutionary changes primarily in very large, freely interbreeding populations. [1]

[Moderate]

Q11. Industrial melanism observed in the peppered moth (*Biston betularia*) during the industrial revolution in England serves as excellent evidence supporting Lamarck's theory of use and disuse of organs. [Hard] [1]

Section B

Answer the following short answer questions. Each question carries 2 marks.

Q12. State the Law of Segregation. Briefly explain why this law is considered universal and without any biological exceptions in sexually reproducing diploid organisms. [Moderate] **[2]**

Q13. Identify and list any four major evolutionary forces or factors that are known to disturb the genetic equilibrium of a population under the Hardy-Weinberg Principle. [Moderate] **[2]**

Q14. Differentiate clearly between 'Incomplete Dominance' and 'Codominance' by providing one distinct biological example for each genetic phenomenon. [Hard] **[2]**

Section C

Answer the following long answer questions. Provide detailed explanations and diagrams/schemes where necessary. Each question carries 4 marks.

Q15. A homozygous tall plant bearing round seeds (TTRR) is crossed with a dwarf plant bearing wrinkled seeds (ttrr). Using a detailed Punnett square scheme, depict the F1 and F2 generations. Derive and state the phenotypic ratio and genotypic ratio of the F2 generation. [4]

[illegible]

Q16. Explain the concept of Natural Selection with reference to the rapid development of drug-resistant microbes. Why is this phenomenon classified as an evolutionary process driven by anthropogenic actions? [4]

[Hard]

— End of Question Paper —